Configuration

Manim provides an extensive configuration system that allows it to adapt to many different use cases. There are many configuration options that can be configured at different times during the scene rendering process. Each option can be configured programmatically via the ManimConfig class, at the time of command invocation via command-line arguments, or at the time the library is first imported via the config files.

The most common, simplest, and recommended way to configure Manim is via the command-line interface (CLI), which is described directly below.

Command-line arguments

By far the most commonly used command in the CLI is the `render` command, which is used to render scene(s) to an output file. It is used with the following arguments:

```
/home/docs/checkouts/readthedocs.org/user_builds/manimce/envs/stable/lib/python3.13/si
  warn("Couldn't find ffmpeg or avconv - defaulting to ffmpeg, but may not work", Runt
Manim Community v0.19.0

Usage: manim render [OPTIONS] FILE [SCENE_NAMES]...

  Render SCENE(S) from the input FILE.

  FILE is the file path of the script or a config file.
...
```

However, since Manim defaults to the `render` command whenever no command is specified, the following form is far more common and can be used instead:

```
manim [OPTIONS] FILE [SCENES]
```

An example of using the above form is:

```
manim -qm file.py SceneOne
```

This asks Manim to search for a Scene class called `SceneOne` inside the file `file.py` and render it with medium quality (specified by the `-qm` flag).

Another frequently used flag is `-p` (“preview”), which makes manim open the rendered video after rendering.

> **Note**
>
> The `-p` flag does not change any properties of the global `config` dict. The `-p` flag is only a command-line convenience.

Advanced examples

To render a scene in high quality, but only output the last frame of the scene instead of the whole video, you can execute

```
manim -sqh <file.py> SceneName
```

The following example specifies the output file name (with the `-o` flag), renders only the first ten animations (`-n` flag) with a white background (`-c` flag), and saves the animation as a `.gif` instead of as a `.mp4` file (`--format=gif` flag). It uses the default quality and does not try to open the file after it is rendered.

```
manim -o myscene --format=gif -n 0,10 -c WHITE <file.py> SceneName
```

A list of all CLI flags

```
$ manim --help
/home/docs/checkouts/readthedocs.org/user_builds/manimce/envs/stable/lib/python3.13/si
  warn("Couldn't find ffmpeg or avconv - defaulting to ffmpeg, but may not work", Runt
Usage: manim [OPTIONS] COMMAND [ARGS]...

  Animation engine for explanatory math videos.

Options:
  --version                      Show version and exit.
  --show-splash / --hide-splash  Print splash message with version information.
  --help                         Show this message and exit.

Commands:
  cfg          Manages Manim configuration files.
  checkhealth  This subcommand checks whether Manim is installed correctly...
  init         Create a new project or insert a new scene.
  plugins      Manages Manim plugins.
  render       Render SCENE(S) from the input FILE.

See 'manim <command>' to read about a specific subcommand.

Note: the subcommand 'manim render' is called if no other subcommand is
specified. Run 'manim render --help' if you would like to know what the '-ql' or
'-p' flags do, for example.
```

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```
$ manim render --help
/home/docs/checkouts/readthedocs.org/user_builds/manimce/envs/stable/lib/python3.13/si
  warn("Couldn't find ffmpeg or avconv - defaulting to ffmpeg, but may not work", Runt
Manim Community v0.19.0

Usage: manim render [OPTIONS] FILE [SCENE_NAMES]...

  Render SCENE(S) from the input FILE.

  FILE is the file path of the script or a config file.

  SCENES is an optional list of scenes in the file.

Global options:
  -c, --config_file TEXT         Specify the configuration file to use for
                                 render settings.
  --custom_folders               Use the folders defined in the [custom_folders]
                                 section of the config file to define the output
                                 folder structure.
  --disable_caching              Disable the use of the cache (still generates
                                 cache files).
  --flush_cache                  Remove cached partial movie files.
  --tex_template TEXT            Specify a custom TeX template file.
  -v, --verbosity [DEBUG|INFO|WARNING|ERROR|CRITICAL]
                                 Verbosity of CLI output. Changes ffmpeg log
                                 level unless 5+.
  --notify_outdated_version / --silent
                                 Display warnings for outdated installation.
  --enable_gui                   Enable GUI interaction.
  --gui_location TEXT            Starting location for the GUI.
  --fullscreen                   Expand the window to its maximum possible size.
  --enable_wireframe             Enable wireframe debugging mode in opengl.
  --force_window                 Force window to open when using the opengl
                                 renderer, intended for debugging as it may
                                 impact performance
  --dry_run                      Renders animations without outputting image or
                                 video files and disables the window
  --no_latex_cleanup             Prevents deletion of .aux, .dvi, and .log files
                                 produced by Tex and MathTex.
  --preview_command TEXT         The command used to preview the output file
                                 (for example vlc for video files)

Output options:
  -o, --output_file TEXT         Specify the filename(s) of the rendered
                                 scene(s).
  -0, --zero_pad INTEGER RANGE   Zero padding for PNG file names.  [0<=x<=9]
  --write_to_movie               Write the video rendered with opengl to a file.
  --media_dir PATH               Path to store rendered videos and latex.
  --log_dir PATH                 Path to store render logs
  --log_to_file                  Log terminal output to fi
```



```
Render Options:
  -n, --from_animation_number TEXT
```

```
                                Start rendering from n_0 until n_1. If n_1 is
                                left unspecified, renders all scenes after n_0.
  -a, --write_all               Render all scenes in the input file.
  --format [png|gif|mp4|webm|mov]
  -s, --save_last_frame         Render and save only the last frame of a scene
                                as a PNG image.
  -q, --quality [l|m|h|p|k]     Render quality at the follow resolution
                                framerates, respectively: 854x480 15FPS,
                                1280x720 30FPS, 1920x1080 60FPS, 2560x1440
                                60FPS, 3840x2160 60FPS
  -r, --resolution TEXT         Resolution in "W,H" for when 16:9 aspect ratio
                                isn't possible.
  --fps, --frame_rate FLOAT     Render at this frame rate.
  --renderer [cairo|opengl]     Select a renderer for your Scene.
  -g, --save_pngs               Save each frame as png (Deprecated).
  -i, --save_as_gif             Save as a gif (Deprecated).
  --save_sections               Save section videos in addition to movie file.
  -t, --transparent             Render scenes with alpha channel.
  --use_projection_fill_shaders Use shaders for OpenGLVMobject fill which are
                                compatible with transformation matrices.
  --use_projection_stroke_shaders
                                Use shaders for OpenGLVMobject stroke which are
                                compatible with transformation matrices.

Ease of access options:
  --progress_bar [display|leave|none]
                                Display progress bars and/or keep them
                                displayed.
  -p, --preview                 Preview the Scene's animation. OpenGL does a
                                live preview in a popup window. Cairo opens the
                                rendered video file in the system default media
                                player.
  -f, --show_in_file_browser    Show the output file in the file browser.
  --jupyter                     Using jupyter notebook magic.

Other options:
  --help                        Show this message and exit.

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```

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```
$ manim cfg --help
/home/docs/checkouts/readthedocs.org/user_builds/manimce/envs/stable/lib/python3.13/si
  warn("Couldn't find ffmpeg or avconv - defaulting to ffmpeg, but may not work", Runt
Manim Community v0.19.0

Usage: manim cfg [OPTIONS] COMMAND [ARGS]...

  Manages Manim configuration files.

Options:
  --help  Show this message and exit.

Commands:
  export
  show
  write

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```

```
$ manim plugins --help
/home/docs/checkouts/readthedocs.org/user_builds/manimce/envs/stable/lib/python3.13/si
  warn("Couldn't find ffmpeg or avconv - defaulting to ffmpeg, but may not work", Runt
Manim Community v0.19.0

Usage: manim plugins [OPTIONS]

  Manages Manim plugins.

Options:
  -l, --list  List available plugins.
  --help      Show this message and exit.

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```

The ManimConfig class

The most direct way of configuring Manim is through the global `config` object, which is an instance of `ManimConfig`. Each property of this class is a config option that can be accessed either with standard attribute syntax or with dict-like syntax:

```
>>> from manim import *
>>> config.background_color = WHITE
>>> config["background_color"] = WHITE
```


…red; the latter is provided for backwards compatibility.

Most classes, including `Camera`, `Mobject`, and `Animation`, read some of their default configuration from the global `config`.

```
>>> Camera({}).background_color
<Color white>
>>> config.background_color = RED  # 0xfc6255
>>> Camera({}).background_color
<Color #fc6255>
```

`ManimConfig` is designed to keep internal consistency. For example, setting `frame_y_radius` will affect `frame_height`:

```
>>> config.frame_height
8.0
>>> config.frame_y_radius = 5.0
>>> config.frame_height
10.0
```

The global `config` object is meant to be the single source of truth for all config options. All of the other ways of setting config options ultimately change the values of the global `config` object.

The following example illustrates the video resolution chosen for examples rendered in our documentation with a reference frame.

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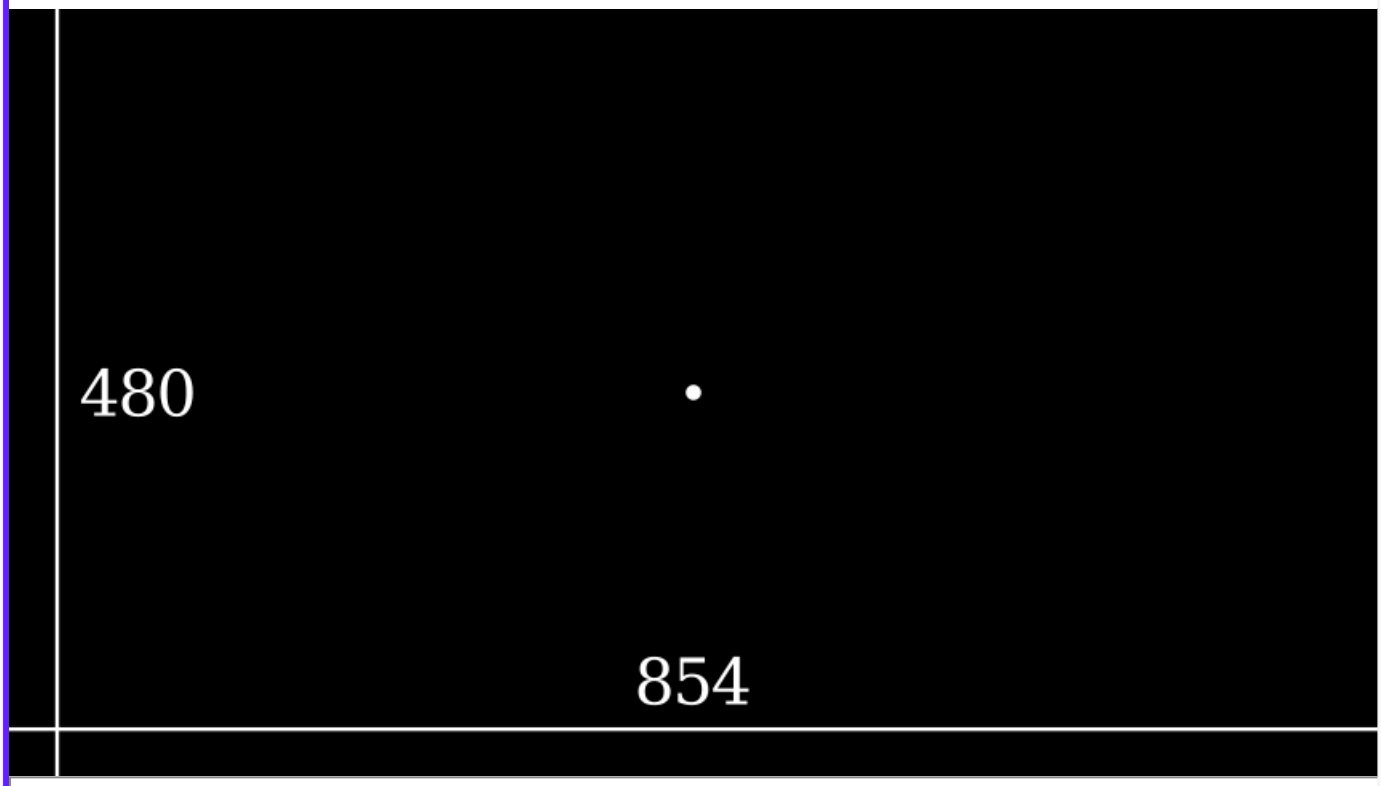
Example: ShowScreenResolution




```
from manim import *

class ShowScreenResolution(Scene):
    def construct(self):
        pixel_height = config["pixel_height"]  #  1080 is default
        pixel_width = config["pixel_width"]  # 1920 is default
        frame_width = config["frame_width"]
        frame_height = config["frame_height"]
        self.add(Dot())
        d1 = Line(frame_width * LEFT / 2, frame_width * RIGHT / 2).to_edge
        self.add(d1)
        self.add(Text(str(pixel_width)).next_to(d1, UP))
        d2 = Line(frame_height * UP / 2, frame_height * DOWN / 2).to_edge(
        self.add(d2)
        self.add(Text(str(pixel_height)).next_to(d2, RIGHT))
```


```
class ShowScreenResolution(Scene):
    def construct(self):
        pixel_height = config["pixel_height"]  #  1080 is default
        pixel_width = config["pixel_width"]  # 1920 is default
        frame_width = config["frame_width"]
        frame_height = config["frame_height"]
        self.add(Dot())
        d1 = Line(frame_width * LEFT / 2, frame_width * RIGHT / 2).to_edge(DOWN)
        self.add(d1)
               '(Text(str(pixel_width)).next_to(d1, UP))
               e(frame_height * UP / 2, frame_height * DOWN
               .add(d2)
        self.add(Text(str(pixel_height)).next_to(d2, RIGHT))
```

The config files

As the last example shows, executing Manim from the command line may involve using many flags simultaneously. This may become a nuisance if you must execute the same script many times in a short time period, for example, when making small incremental tweaks to your scene script. For this reason, Manim can also be configured using a configuration file. A configuration file is a file ending with the suffix `.cfg`.

To use a local configuration file when rendering your scene, you must create a file with the name `manim.cfg` in the same directory as your scene code.

> **Warning**
>
> The config file **must** be named `manim.cfg`. Currently, Manim does not support config files with any other name.

The config file must start with the section header `[CLI]`. The configuration options under this header have the same name as the CLI flags and serve the same purpose. Take, for example, the following config file.

```
[CLI]
# my config file
output_file = myscene
save_as_gif = True
background_color = WHITE
```

Config files are parsed with the standard python library `configparser`. In particular, they will ignore any line that starts with a pound symbol `#`.

Now, executing the following command

```
manim -o myscene -i -c WHITE <file.py> SceneName
```

is equivalent to executing the following command, provided that `manim.cfg` is in the same directory as <file.py>,

```
manim <file.py> SceneName
```

> **Tip**
>
> The names of the configuration options admissible in config files are exactly the same as the **long names** of the corresponding command- line flags. For example, the `-c` and `--background_color` flags are interchangeable, but the config file only accepts `background_color` as an admissible option.

 are meant to replace CLI flags, all CLI flags can moreover, any config option can be set via a config file, whether or not it has an associated CLI flag. See the bottom of this document for a list of all CLI flags and config options.

Manim will look for a `manim.cfg` config file in the same directory as the file being rendered, and **not** in the directory of execution. For example,

```
manim -o myscene -i -c WHITE <path/to/file.py> SceneName
```

will use the config file found in `path/to/file.py`, if any. It will **not** use the config file found in the current working directory, even if it exists. In this way, the user may keep different config files for different scenes or projects, and execute them with the right configuration from anywhere in the system.

The file described here is called the **folder-wide** config file because it affects all scene scripts found in the same folder.

The user config file

As explained in the previous section, a `manim.cfg` config file only affects the scene scripts in its same folder. However, the user may also create a special config file that will apply to all scenes rendered by that user. This is referred to as the **user-wide** config file, and it will apply regardless of where Manim is executed from, and regardless of where the scene script is stored.

The user-wide config file lives in a special folder, depending on the operating system.

- Windows: `UserDirectory` /AppData/Roaming/Manim/manim.cfg
- MacOS: `UserDirectory` /.config/manim/manim.cfg
- Linux: `UserDirectory` /.config/manim/manim.cfg

Here, `UserDirectory` is the user's home folder.

> **Note**
>
> A user may have many **folder-wide** config files, one per folder, but only one **user-wide** config file. Different users in the same computer may each have their own user-wide config file.

> **Warning**
>
> Do not store scene scripts in the same folder as the user-wide config file. In this case, the behavior is undefined.

Whenever you use Manim from anywhere in the system, Manim will look for a user-wide config file and read its configuration.

Cascading config files

What happens if you execute Manim and it finds both a folder-wide config file and a user-wide

 will read both files, but if they are incompatible

.

For example, take the following user-wide config file

```
# user-wide
[CLI]
output_file = myscene
save_as_gif = True
background_color = WHITE
```

and the following folder-wide file

```
# folder-wide
[CLI]
save_as_gif = False
```

Then, executing `manim <file.py> SceneName` will be equivalent to not using any config files and executing

```
manim -o myscene -c WHITE <file.py> SceneName
```

Any command-line flags have precedence over any config file. For example, using the previous two config files and executing `manim -c RED <file.py> SceneName` is equivalent to not using any config files and executing

```
manim -o myscene -c RED <file.py> SceneName
```

There is also a **library-wide** config file that determines Manim's default behavior and applies to every user of the library. It has the least precedence, so any config options in the user-wide and any folder-wide files will override the library-wide file. This is referred to as the *cascading* config file system.

> Warning
>
> **The user should not try to modify the library-wide file**. Contributors should receive explicit confirmation from the core developer team before modifying it.

Order of operations

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